

工學學士 學位論文

Context Switching Performance Analysis in Rust-based RTOS with MPU

(ARM Cortex-M4에서 MPU를 적용한 Rust 기반 RTOS의 Context
Switching 성능 분석)

2025년 11월

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조용휘

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Switching 성능 분석)

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이 논문을 공학 학사학위 논문으로 제출함.

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조용휘

조용휘의 공학 학사학위 논문을 인준함.

2025년 11월 25일

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Abstract

This paper presents a comprehensive safety and security evaluation of task-level memory isolation on ARM Cortex-M4-based embedded systems, aimed at automotive, industrial, and IoT safety-critical applications. Ensuring functional safety (ISO 26262) and cybersecurity requires strict separation between software components, yet traditional process isolation is not available on microcontrollers. To address this gap, we implement an MPU-based task isolation framework within a lightweight Rust RTOS, providing strong inter-task separation, memory-violation detection, and deterministic real-time behavior. The proposed system combines privilege-level execution with Memory Protection Unit enforcement to guarantee that each task operates within its assigned memory region. Isolation correctness is experimentally validated using adversarial tasks attempting unauthorized cross-memory access. All violations are successfully detected and contained through hardware MemManage faults, confirming that tasks cannot interfere with or corrupt each other's state. This demonstrates that complete task-to-task isolation is achievable even on resource-constrained MCUs. To assess real-time feasibility, cycle-accurate DWT measurements quantify both the baseline context switching cost and MPU-related overheads. MPU-enabled context switching adds only **20.7 ± 2.0 cycles (1.25%)**, while MPU region configuration incurs a deterministic **994 cycle (5.522 μs)** cost per protected task, scaling linearly with the number of active IPC applications. In systems with 2–3 tasks, total isolation overhead remains within **11–16.6 μs**, representing < 0.5% of a typical 1 ms automotive control cycle. These results confirm that the required isolation does not compromise real-time responsiveness. Our findings establish that strong safety and security properties—including fault containment, temporal determinism, and memory access control—can be implemented on low-power microcontrollers without sacrificing performance. The demonstrated framework provides a practical foundation for secure, multi-tasking embedded architectures, enabling ISO 26262-compliant functional safety and modern cybersecurity capabilities in automotive ECUs, industrial controllers, and constrained IoT devices.

Keywords : ARM Cortex-M4, Memory Protection Unit, Real-time Operating Systems, Context Switching

Student ID : 2018-19609

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Chapter I Introduction

Modern automotive systems are experiencing a fundamental transformation toward software-defined vehicles, where Electronic Control Units (ECUs) assume increasingly critical roles in vehicle operation, safety, and user experience. Contemporary vehicles integrate dozens of ECUs that control functions ranging from engine management to autonomous driving capabilities. As these systems grow in complexity, ensuring their reliability, security, and real-time performance becomes paramount for automotive safety and functionality.

ARM Cortex-M4 microcontrollers, particularly the STM32F446 series, have gained widespread adoption in automotive ECU applications due to their optimal balance of performance, power efficiency, and cost-effectiveness. The STM32F446 operates at 180MHz, delivering 225 DMIPS performance while maintaining low power consumption suitable for automotive environments. These characteristics make it an ideal platform for implementing safety-critical automotive functions that require both computational capability and energy efficiency.

One distinguishing feature of Cortex-M4 processors for safety-critical applications is the optional Memory Protection Unit (MPU). The MPU provides hardware-enforced memory access control, enabling implementation of memory isolation between different software components. This capability is essential for meeting automotive functional safety standards such as ISO 26262, which mandates robust fault detection and containment mechanisms to prevent catastrophic system failures.

However, introducing memory protection comes with performance implications that must be carefully considered. Every memory access requires validation against MPU configurations, and context switches between protected domains necessitate additional overhead for reconfiguring memory regions. Understanding these performance characteristics is crucial for automotive system designers who must balance safety requirements with stringent real-time constraints.

Real-Time Operating Systems (RTOS) in automotive applications must guarantee deterministic behavior with strict timing requirements. Engine control systems require sub-millisecond response times for critical operations, while Anti-lock Braking Systems (ABS) and Electronic Stability Control (ESC) have even tighter constraints. The performance overhead introduced by MPU protection could potentially violate these timing requirements if not properly characterized and optimized, making this research essential for automotive safety.

1.1 Research Motivation

The automotive industry's shift toward higher levels of automation and connectivity has dramatically increased the security and safety requirements for ECU systems. Traditional automotive software architectures, which relied primarily on software-based isolation mechanisms, are insufficient for meeting modern security threats and functional safety standards.

Hardware-based memory protection through MPU implementation offers a robust solution, but its performance impact on real-time automotive applications remains poorly understood.

Current literature lacks comprehensive performance analysis of MPU-enabled systems specifically tailored for automotive ECU environments. While general-purpose embedded system studies exist, automotive applications have unique characteristics including strict real-time constraints, specific workload patterns, and safety-critical requirements that demand specialized analysis. This research gap motivates the need for detailed performance characterization of MPU overhead in automotive contexts.

Furthermore, the emergence of memory-safe programming languages like Rust in automotive development presents an opportunity to complement hardware-based protection with compile-time safety guarantees. Understanding how these technologies interact and their combined impact on system performance is essential for developing next-generation automotive software architectures.

1.2 Research Objectives

This thesis aims to provide comprehensive performance analysis of MPU-enabled systems in automotive ECU environments. The primary research objectives include:

1. **Quantitative Performance Characterization:** Measure and analyze precise performance overhead introduced by MPU protection across various system operations including context switching, inter-process communication, and memory access patterns using cycle-accurate measurement techniques.
2. **Real-time Constraint Validation:** Evaluate whether MPU-protected systems can meet typical automotive ECU timing requirements across different application categories and identify potential performance bottlenecks that could impact safety-critical operations.
3. **Optimization Strategy Development:** Develop and validate techniques for minimizing MPU overhead while maintaining security and safety guarantees, focusing on practical solutions applicable to production automotive systems.
4. **Implementation Guidelines:** Provide actionable recommendations for automotive ECU designers implementing MPU-based memory protection, including configuration strategies and performance optimization techniques.

The research focuses specifically on the STM32F446 microcontroller running a custom Rust-based RTOS, representing a modern approach to automotive software development that emphasizes both memory safety and performance optimization.

1.3 Research Contributions

This work makes several significant contributions to the field of automotive embedded systems and real-time system design:

Comprehensive MPU Performance Analysis: First detailed study of MPU over-head specifically for automotive ECU applications, providing cycle-accurate measurements across multiple system operations and configurations.

Optimization Techniques: Development and validation of systematic optimization strategies that reduce MPU overhead by up to 65% while maintaining security properties, making hardware protection practically feasible for automotive applications.

Real-time Constraint Validation: Demonstration that MPU-protected systems can meet automotive timing requirements across various ECU categories, supporting the adoption of hardware-based security in safety-critical automotive applications.

Implementation Reference: Production-ready Rust RTOS implementation with MPU support, serving as a reference for automotive developers seeking to implement hardware-based memory protection in their systems.

1.4 Thesis Organization

This thesis is organized into six chapters that systematically address the research objectives and present comprehensive analysis of MPU performance in automotive ECU environments.

Chapter 2 provides essential background information on ARM Cortex-M4 architecture, MPU functionality, automotive RTOS requirements, and existing approaches to embedded system performance analysis. This foundation enables understanding of the technical challenges and opportunities in automotive ECU design.

Chapter 3 describes the system design and implementation details of the evaluation platform, including the STM32F446 hardware configuration, Rust-based RTOS architecture, MPU configuration strategies, and performance measurement infrastructure used throughout the research.

Chapter 4 outlines the experimental methodology and benchmark design, covering the comprehensive test scenarios developed to characterize MPU performance across context switching, inter-process communication, and memory access operations representative of automotive workloads.

Chapter ?? presents detailed results and analysis of performance measurements, including quantitative characterization of MPU overhead, statistical analysis of timing variations, and validation of real-time constraint compatibility across different automotive ECU categories.

Chapter ?? discusses optimization strategies and trade-offs between security and performance, providing practical techniques for minimizing MPU overhead while maintaining safety guarantees required for automotive applications.

Finally, Chapter ?? concludes with a summary of contributions, discussion of limitations, and suggestions for future research directions that could extend this work to broader automotive system contexts and emerging technologies.

Chapter 2

Background and Related Work

This chapter provides essential background information and reviews related work in areas fundamental to understanding MPU performance analysis in automotive ECU environments. We begin with an overview of ARM Cortex-M4 architecture and Memory Protection Unit functionality, followed by automotive RTOS requirements and existing performance measurement methodologies.

2.1 ARM Cortex-M4 Architecture

The ARM Cortex-M4 processor is a 32-bit RISC processor designed specifically for embedded applications requiring high performance and energy efficiency. The architecture features a 3-stage pipeline with Harvard memory architecture, enabling simultaneous instruction fetch and data access operations that contribute to its performance characteristics in real-time applications [5].

2.1.1 Core Architecture Features

Key architectural features relevant to this research include:

Instruction Set Architecture: The Cortex-M4 implements the ARMv7-M architecture with Thumb-2 instruction set, providing 32-bit performance with 16-bit instruction density. This design enables efficient code density while maintaining high performance, crucial for automotive ECU applications with limited memory resources.

Pipeline Organization: The 3-stage pipeline consists of fetch, decode, and execute stages. While simpler than higher-performance ARM cores, this design provides predictable timing behavior essential for real-time automotive applications where worst-case execution time analysis is critical.

Memory System: Harvard architecture with separate instruction and data buses enables simultaneous instruction fetch and data access. The architecture supports configurable memory regions with different access characteristics, directly relevant to MPU functionality and performance analysis.

2.1.2 Performance Characteristics

The STM32F446 implementation operates at up to 180MHz, providing 225 DMIPS peak performance. Key performance features include:

- Single-cycle multiply instructions for efficient mathematical computations common in automotive control algorithms

- Hardware divide instructions reducing software overhead for division operations

- Optional floating-point unit (FPU) supporting single-precision operations required for automotive sensor data processing

- Advanced High-performance Bus (AHB) architecture enabling efficient peripheral access

2.1.3 Debug and Trace Capabilities

The Cortex-M4 includes comprehensive debug and trace capabilities essential for performance analysis:

- Data Watchpoint and Trace (DWT) Unit:** Provides cycle-accurate performance monitoring through hardware counters. The DWT CYCCNT register increments with each processor cycle, enabling precise execution time measurement down to 5.56 nanoseconds at 180MHz operation.

- Instrumentation Trace Macrocell (ITM):** Supports real-time debugging and performance profiling through non-intrusive trace data collection, crucial for analyzing real-time system behavior without disturbing timing characteristics.

2.2 Memory Protection Unit Overview

The ARM Cortex-M4 MPU is an optional component providing hardware-enforced memory access control. Understanding MPU functionality and configuration options is fundamental to analyzing its performance impact in automotive applications.

2.2.1 MPU Architecture and Configuration

The STM32F446 implementation includes an 8-region MPU supporting the following protection attributes:

Memory Region Configuration:

- Up to 8 simultaneously active memory regions with independent attribute configuration

- Minimum region size of 32 bytes with power-of-2 sizing up to 4GB

- Configurable base addresses that must be aligned to region size boundaries

- Sub-region disable functionality for fine-grained access control within larger regions

Access Control Attributes:

Privileged and unprivileged access permissions for implementing privilege separation

Read, write, and execute permission combinations enabling data execution prevention

Shareable and cacheable memory attributes for optimizing memory system performance

Memory ordering guarantees (strongly ordered, device, normal memory types)

2.2.2 Protection Mechanisms

When enabled, the MPU validates every memory access against configured regions using hardware logic that operates in parallel with normal memory operations. This validation process introduces latency that varies based on several factors:

Access Validation Process: Each memory access triggers hardware comparison against active region descriptors. Multiple regions may be checked in parallel, but the validation process adds cycles to memory access operations.

Fault Handling: Access violations trigger Memory Management Fault exceptions, enabling system-level fault handling policies. The fault handling mechanism provides diagnostic information about violation types and addresses, supporting both security and safety requirements.

Dynamic Reconfiguration: MPU regions can be reconfigured during runtime, particularly during context switches. This capability is essential for task isolation but introduces overhead that must be characterized for real-time applications.

2.3 Automotive ECU RTOS Requirements

Automotive ECUs operate under stringent constraints that combine real-time performance requirements with functional safety standards. Understanding these requirements is essential for evaluating MPU performance impact in automotive contexts.

2.3.1 Real-time Performance Requirements

Automotive applications exhibit diverse timing requirements based on their safety criticality and functional domains:

Engine Control Systems: Require sub-millisecond response times for fuel injection timing and ignition control. These systems typically operate with 1-5ms control loop periods and must maintain deterministic behavior under all operating conditions.

Chassis Control Systems: ABS, ESC, and traction control systems demand even tighter timing constraints, often requiring response times under 10ms for critical safety interventions. **Body Electronics:** Door controls, lighting systems, and comfort functions operate with more relaxed timing requirements, typically 50-100ms response times, but must still maintain predictable behavior.

2.3.2 Functional Safety Standards

ISO 26262 defines functional safety requirements for automotive systems, establishing Auto-motive Safety Integrity Levels (ASIL) that directly impact system design:

ASIL Classification: Ranges from ASIL A (lowest integrity) to ASIL D (highest integrity), with higher levels requiring more stringent fault detection and mitigation mechanisms.

Memory Protection Requirements: ASIL C and ASIL D systems typically require hardware-based memory protection to ensure fault isolation and prevent error propagation between safety-critical and non-critical functions.

Diagnostic Coverage: Higher ASIL levels mandate diagnostic coverage requirements (90% for ASIL C/D), making hardware-based fault detection mechanisms like MPU violation detection valuable for safety certification.

2.3.3 Security Requirements

Modern automotive systems face increasing cybersecurity threats requiring robust protection mechanisms:

Attack Surface Mitigation: Connected vehicles present numerous attack vectors through wireless interfaces, diagnostic ports, and network communications. Memory protection helps limit attack impact by containing compromised software components.

Code Integrity: Protection against code injection attacks and return-oriented programming requires hardware enforcement of execute permissions and control flow integrity.

2.4 Performance Measurement Methodologies

Accurate performance measurement in embedded systems requires specialized techniques that minimize measurement perturbation while providing sufficient precision for real-time analysis.

2.4.1 Cycle-Level Measurement Techniques

Hardware performance counters provide the most accurate measurement capability for embedded systems:

DWT Cycle Counter: The ARM Cortex-M4 DWT unit provides zero-overhead cycle counting through dedicated hardware registers. The 32-bit cycle counter wraps every 2^{32} cycles (approximately 24 seconds at 180MHz), requiring careful handling for long-duration measurements.

Measurement Precision: Cycle-level measurement enables analysis of timing variations at the instruction level, essential for understanding MPU overhead characteristics and their impact on real-time constraints.

2.4.2 Statistical Analysis for Real-time Systems

Real-time system performance exhibits statistical variation due to multiple factors:

Sources of Variation: Pipeline effects, memory system latencies, interrupt timing, and cache behavior contribute to execution time variability that must be characterized statistically. **Appropriate Metrics:** Real-time systems require analysis of worst-case execution times, timing distribution characteristics, and jitter analysis rather than simple average performance measurements.

2.5 Related Work

Previous research in embedded system performance analysis and memory protection provides foundation for this automotive-specific study.

2.5.1 Embedded System Performance Analysis

Liu and Layland [8] established fundamental real-time scheduling theory that remains relevant for automotive ECU design. Their rate monotonic analysis provides mathematical framework for determining system schedulability under timing constraints.

Buttazzo [9] comprehensively covers hard real-time computing systems, including performance analysis techniques applicable to automotive embedded systems. The work emphasizes importance of worst-case execution time analysis for safety-critical applications.

2.5.2 Memory Protection in Embedded Systems

Recent work by Kleidermacher and Kleidermacher [19] addresses embedded systems security, including memory protection mechanisms. However, their focus on general embedded applications lacks automotive-specific performance analysis.

Anderson [20] provides comprehensive coverage of security engineering principles applicable to automotive systems, but does not address specific performance implications of hardware protection mechanisms in real-time contexts.

2.5.3 Automotive Software Architecture

The AUTOSAR consortium [4] has established industry standards for automotive software architecture, including recommendations for memory protection usage. However, detailed performance analysis of these recommendations remains limited in public literature.

Recent automotive cybersecurity research [11, 12] demonstrates critical need for robust isolation mechanisms in automotive systems, but focuses primarily on attack methodologies rather than protection system performance characteristics.

This research addresses the gap in automotive-specific MPU performance analysis, providing detailed characterization necessary for implementing hardware-based protection in real-time automotive applications.

Chapter 3

System Design and Implementation

This chapter describes the system design and implementation of the evaluation platform used to analyze MPU performance in automotive ECU environments. The platform consists of STM32F446 hardware, custom Rust-based RTOS implementation, MPU configuration management, and performance measurement infrastructure designed for cycle-accurate analysis.

3.1 Hardware Platform: STM32F446

The STM32F446RE microcontroller serves as the evaluation platform, representing typical automotive-grade Cortex-M4 implementations. This device provides representative performance characteristics and MPU functionality found in production automotive ECU systems.

3.1.1 Core Specifications

The STM32F446RE provides the following specifications relevant to automotive ECU applications:

- ARM Cortex-M4F processor with single-precision FPU
- 180MHz maximum operating frequency providing 225 DMIPS performance
- 512KB Flash memory for application code and configuration data
- 128KB SRAM for runtime data and task stacks
- 8-region Memory Protection Unit with sub-region disable capability

3.1.2 Automotive-Relevant Features

The STM32F446 includes features specifically designed for automotive environments:

Operating Conditions: Extended temperature range (-40°C to +105°C) and high electromagnetic compatibility (EMC) robustness ensure reliable operation in harsh automotive environments.

Communication Interfaces: CAN bus interface enables integration with automotive networks, while multiple UART, SPI, and I2C interfaces support sensor and actuator communication typical in automotive applications.

Power Management: Advanced power management capabilities including multiple low-power modes support automotive power budget requirements and enable efficient operation during vehicle standby states.

3.1.3 Development Platform Configuration

The evaluation uses an STM32F446RE Nucleo development board configured for representative automotive operation:

Clock Configuration: The system operates at maximum 180MHz frequency using an external high-speed crystal oscillator, ensuring timing accuracy required for cycle-precise measurements.

Memory Layout: The memory configuration follows automotive ECU conventions with clear separation between program code, runtime data, and peripheral memory regions to enable meaningful MPU protection analysis.

3.2 Rust-based RTOS Implementation

This research implements a custom RTOS in Rust, a systems programming language offering memory safety guarantees that complement hardware-based protection. The RTOS design follows automotive best practices while enabling detailed performance analysis.

3.2.1 Language Choice Rationale

Rust provides several advantages for automotive ECU development that justify its selection for this research:

Memory Safety: Compile-time memory safety verification eliminates entire classes of vulnerabilities, complementing hardware MPU protection with software-level guarantees.

Zero-Cost Abstractions: Rust's abstraction mechanisms compile to efficient machine code without runtime overhead, essential for meeting automotive performance requirements.

Deterministic Behavior: Absence of garbage collection and predictable memory allocation patterns support real-time automotive applications requiring deterministic timing behavior.

3.2.2 RTOS Architecture

The RTOS implements essential functionality required for automotive ECU operation:

Task Management: The task management system implements priority-based preemptive scheduling with round-robin scheduling within priority levels. Each task includes metadata for MPU configuration, enabling automatic memory protection setup during context switches.

Memory Management: Static memory allocation eliminates heap fragmentation and provides deterministic memory access patterns. Task stacks are allocated at compile time with

Result: Task scheduling and MPU management

Function `schedule(current task)`:

```
next task ← priority queue.pop();  
if next task.mpu config ≠ current task.mpu config then  
    configure_mpu(next task.mpu config);  
end  
context_switch(current task, next task);
```

Algorithm 1: Task Scheduling with MPU Management

MPU guard regions to detect stack overflow conditions.

Synchronization Primitives: The RTOS provides mutexes, semaphores, and message queues with priority inheritance support to prevent priority inversion in safety-critical automotive applications.

3.2.3 Hardware Abstraction Layer

The hardware abstraction layer (HAL) provides safe interfaces to STM32F446 peripherals while enabling performance measurement:

Listing 3.1: MPU Configuration Interface

```
1 pub struct Mpu Manager {  
2     regions : [ Option < MpuRegion >; 8]  
3     , active_config : Mpu Configuration ,  
4 }  
5  
6 impl Mpu Manager {  
7     pub unsafe fn configure_region ( &  
8         mut self ,  
9         id: RegionId ,  
10        region : Mpu Region  
11    ) -> Result <() , MpuError > {  
12        // Configure MPU region with cycle counting let  
13        start_cycles = dwt :: get_cycles (); self .  
14        write_mpu_registers ( id , & region );  
15        let end_cycles = dwt :: get_cycles ();  
16  
17        // Record configuration overhead  
18        self . record_config_overhead ( end_cycles - start_cycles );  
19        Ok (())  
20    }  
21 }
```

3.3 MPU Configuration and Management

The MPU implementation provides comprehensive memory protection while enabling detailed performance analysis of protection overhead.

3.3.1 Memory Region Strategy

The MPU configuration strategy balances security requirements with performance considerations:

Region 0 - Flash Memory:

Address Range: 0x08000000 - 0x0807FFFF (512KB)

Attributes: Cacheable, Execute/Read permissions

Purpose: Program code protection with execute permission enforcement

Region 1 - System RAM:

Address Range: 0x20000000 - 0x2001FFFF (128KB)

Attributes: Cacheable, Read/Write permissions (no execute)

Purpose: Data memory with execute prevention for security

Region 2 - Peripheral Memory:

Address Range: 0x40000000 - 0x5FFFFFFF (512MB)

Attributes: Device memory, Read/Write permissions

Purpose: Hardware register access with appropriate ordering guarantees

Regions 3-7 - Task-Specific Protection: Dynamic regions configured per task for stack isolation and data protection, enabling fine-grained memory isolation between automotive software components.

3.3.2 Dynamic MPU Reconfiguration

Context switches require efficient MPU reconfiguration to maintain protection while minimizing overhead:

Listing 3.2: Context Switch MPU Update

```
1 unsafe fn context_switch_with_mpu ( from_task
2   : & mut Task Context , to_task : & Task
3   Context
4 ) {
5   // Measure context switch timing
6   let start_cycles = dwt :: get_cycles ();
7 }
```

```

8      // Save current task state
9      save_task_registers ( from_task );
10
11     // Update MPU if configuration differs
12     if from_task . mpu_config != to_task . mpu_config {
13         mpu_disable      (); configure_task_regions (&
14             to_task . mpu_config ); mpu_enable ();
15     }
16
17     // Load new task state load_task_registers
18     ( to_task );
19
20     let end_cycles = dwt :: get_cycles ();
21     record_context_switch_time ( end_cycles - start_cycles );
22 }
23

```

3.3.3 Optimization Strategies

Several optimization techniques minimize MPU overhead while maintaining security properties: **Lazy Reconfiguration**: MPU regions are updated only when task memory requirements differ, avoiding unnecessary reconfiguration overhead for tasks with identical memory layouts.

Region Caching: Frequently used MPU configurations are cached to reduce setup time during context switches, particularly beneficial for periodic automotive control tasks.

Batch Configuration: Multiple MPU regions are configured simultaneously during single MPU disable periods, minimizing the time window where memory protection is inactive.

3.4 Performance Measurement Infrastructure

Accurate performance measurement requires specialized infrastructure that provides cycle-accurate timing without introducing measurement artifacts.

3.4.1 DWT-based Cycle Counting

The measurement infrastructure utilizes the ARM Cortex-M4 DWT unit for precise timing:

Listing 3.3: DWT Cycle Counter Implementation

```

1  mod dwt {
2      const DWT_CTRL : * mut u32 = 0 x E 0001000 as * mut u32 ;
3      const DWT_CYCCNT : * mut u32 = 0 x E 0001004 as * mut u32 ;
4
5      pub unsafe fn init () {
6          // Enable DWT cycle counter
7          let mut ctrl = ptr :: read_volatile ( DWT_CTRL );

```

```

8      ctrl |= 0 x1 ; // CYCCNTENA bit
9      ptr :: write_volatile ( DWT_CTRL , ctrl );
10
11      // Reset counter to zero
12      ptr :: write_volatile ( DWT_CYCCNT , 0);
13  }
14
15  #[ inline ( always )]
16  pub fn get_cycles () -> u32 {
17      unsafe { ptr :: read_volatile ( DWT_CYCCNT ) }
18  }
19
20  pub fn cycles_to_microseconds ( cycles : u32 ) -> f32 {
21      cycles as f32 / 180.0 // 180 MHz operation
22  }
23 }

```

3.4.2 Statistical Analysis Framework

The measurement framework includes comprehensive statistical analysis capabilities:

Listing 3.4: Benchmark Statistics

```

1  pub struct Benchmark Stats {
2      pub samples : Vec <u32 >,
3      pub min_cycles : u32 ,
4      pub max_cycles : u32 ,
5      pub mean_cycles : f64 ,
6
7      pub std_dev : f64 ,
8      pub percentile_95 : u32 ,
9      pub percentile_99 : u32 ,
10 }
11
12 impl Benchmark Stats {
13     pub fn from_measurements ( measurements : &[ u32 ]) -> Self {
14         let mut sorted = measurements . to_vec (); sorted .
15             sort_unstable ();
16
17         let mean = measurements . iter (). sum :: < u32 >() as
18             f64
19             / measurements . len () as f64 ;
20
21         let variance = measurements . iter ()
22             . map (|& x| ( x as f64 - mean ). powi (2) )
23             . sum :: < f64 >() / measurements . len () as
24             f64 ;
25
26         Self {

```

```

24         samples : measurements . to_vec () , min_cycles
25         : sorted [0] ,
26         max_cycles : sorted [ sorted . len () - 1] ,
27         mean_cycles : mean ,
28         std_dev : variance . sqrt () ,
29         percentile_95 : sorted [ sorted . len () * 95 / 100] ,
30         percentile_99 : sorted [ sorted . len () * 99 / 100] ,
31     }
32 }
33 }

```

3.4.3 Data Collection and Analysis

Performance data collection follows rigorous methodology to ensure statistical validity:

Sample Size: Each benchmark collects minimum 100 samples to ensure statistical significance, with larger sample sizes (1000+) for operations with high variance.

Warm-up Procedures: All benchmarks include warm-up phases to stabilize processor caches and pipeline state before measurement collection begins.

Outlier Detection: Statistical analysis includes outlier detection and filtering to identify measurement artifacts that could skew results.

Real-time Logging: Performance data is collected using Real-Time Transfer (RTT) for external analysis, enabling detailed offline analysis without impacting real-time measurement accuracy.

This comprehensive system design provides the foundation for accurate MPU performance analysis in automotive ECU environments, combining representative hardware with optimized software implementation and precise measurement capabilities.

Chapter 4

Experimental Setup and Methodology

This chapter describes the comprehensive experimental methodology designed to characterize MPU performance impact in automotive ECU environments. The experimental setup includes benchmark design, measurement procedures, and statistical analysis techniques that ensure reliable and representative performance data.

4.1 Benchmark Design Principles

The experimental evaluation employs three primary benchmark categories designed to capture MPU performance impact across typical automotive ECU operations. Each benchmark isolates specific performance characteristics while maintaining relevance to real automotive workloads.

4.1.1 Design Objectives

The benchmark design follows several key principles essential for automotive ECU performance analysis:

Isolation of MPU Effects: Benchmarks are designed to isolate MPU overhead from other system factors, enabling precise measurement of protection-related performance impact.

Automotive Relevance: Test scenarios reflect realistic automotive ECU workloads including periodic control tasks, inter-task communication, and memory access patterns typical in automotive applications.

Statistical Rigor: Each benchmark collects sufficient samples for statistical analysis with appropriate warm-up procedures and outlier detection to ensure measurement validity.

Scalability: Benchmarks evaluate performance across different MPU configuration complexities, from simple 2-region setups to complex 8-region configurations representative of safety-critical automotive systems.

4.1.2 Benchmark Categories

Three primary benchmark categories characterize different aspects of MPU performance impact:

1. **Context Switch Benchmarks:** Measure task switching overhead with and without MPU protection, including analysis of MPU reconfiguration costs and their impact on

real-time scheduling.

- 2. **Inter-Process Communication Benchmarks:** Evaluate IPC mechanisms including message queues and shared memory access under MPU protection, relevant to automotive ECU coordination patterns.
- 3. **Memory Access Pattern Benchmarks:** Analyze memory access performance including sequential and random access patterns, memory allocation operations, and cache behavior with MPU validation.

4.2 Context Switch Performance Measurement

Context switching represents one of the most critical RTOS operations in automotive ECUs, occurring hundreds to thousands of times per second depending on application requirements. Accurate measurement of context switch overhead is essential for validating real-time performance.

4.2.1 Test Scenario Design

The context switch benchmark implements a controlled scenario that isolates switching overhead:

Task Configuration: Two identical tasks alternate execution through yield operations, implementing round-robin scheduling with equal priority to ensure consistent switching behavior.

Workload Minimization: Tasks perform minimal computation to isolate context switching overhead from application-specific processing time.

MPU Configuration Variants: The benchmark evaluates multiple MPU configurations to understand scaling behavior:

Table 4.1: Context Switch Benchmark Configurations

Configuration	Regions	Stack Size	Description
Baseline	0	1024B	No MPU protection
Simple	2	1024B	Basic task isolation
Standard	4	1024B	Typical automotive configuration
Complex	6	1024B	Safety-critical configuration
Maximum	8	1024B	Full MPU capability

4.2.2 Measurement Implementation

The context switch measurement implementation ensures accurate timing capture:

Result: Context switch timing measurement

Function *measure context switch()*:

```
measurements ← [0; 100];
for i ← 0 to 100 do
    task_yield() // Synchronize measurement start
    start_cycles ← dwt::get_cycles();
    task_yield() // Switch to other task and back
    end_cycles ← dwt::get_cycles(); measurements[i] ←
    end_cycles - start_cycles;
end
record_measurement(measurements);
```

Algorithm 2: Context Switch Timing Measurement

4.2.3 Expected Performance Characteristics

Based on ARM Cortex-M4 architecture analysis and preliminary measurements, expected performance ranges include:

Baseline Context Switch: 260-300 cycles (1.4-1.7 s at 180MHz)

MPU-enabled Context Switch: 330-400 cycles (1.8-2.2 s at 180MHz)

MPU Reconfiguration Overhead: 60-100 cycles per context switch

4.3 Inter-Process Communication Evaluation

IPC mechanisms enable coordination between automotive ECU tasks while maintaining isolation. Performance analysis of IPC under MPU protection is crucial for understanding system-level communication costs.

4.3.1 Message Queue Performance Analysis

Message queues provide asynchronous communication suitable for automotive event-driven architectures:

Round-trip Latency Measurement: The benchmark measures complete message round-trip time including queue operations and task scheduling overhead under MPU protection.

Throughput Analysis: Sustained message throughput measurement evaluates system capacity under continuous communication loads typical in automotive networks.

Queue Overflow Handling: Performance measurement of queue overflow conditions and recovery mechanisms relevant to automotive fault handling requirements.

Listing 4.1: IPC Latency Measurement

```
1 fn measure_ipc_round_trip () -> Benchmark Stats { let
2     mut latencies = Vec :: new ();
3
4     for iteration in 0.. 50 {
5         let start = dwt :: get_cycles ();
```

```

6
7      // Send message through protected queue
8      let message = Benchmark Message :: new ( iteration );
9      message_queue . send ( message ). unwrap ();
10
11     // Receive response message
12     let response = message_queue . receive (). unwrap (); let
13     end = dwt :: get_cycles ();
14
15     latencies . push ( end - start );
16 }
17
18 Benchmark Stats :: from_measurements (& latencies )
19 }

```

4.3.2 Shared Memory Access Patterns

Shared memory provides high-performance communication suitable for large data transfers between automotive tasks:

Protected Memory Allocation: Measurement of memory allocation and deallocation operations under MPU protection, including region setup overhead.

Access Pattern Analysis: Evaluation of different memory access patterns (sequential, random, stride-based) with MPU validation overhead.

Synchronization Overhead: Performance impact of mutex-based synchronization for shared memory access under MPU protection.

4.4 Memory Access Pattern Analysis

Memory access patterns significantly impact MPU performance due to permission checking requirements. Comprehensive analysis across different access patterns provides insight into optimization opportunities.

4.4.1 Sequential Access Measurement

Sequential memory access represents optimal access patterns for most automotive data processing operations:

Listing 4.2: Sequential Memory Access Benchmark

```

1 fn benchmark_sequential_access ( buffer : & mut [ u8 ] ) -> u32 {
2     let start = dwt :: get_cycles ();
3
4     // Write phase - sequential pattern for i
5     in 0.. buffer . len () {
6         buffer [ i ] = ( i & 0xFF ) as u8 ;

```

```

7      }
8
9      // Read phase - sequential pattern with computation let mut
10     checksum = 0 u32 ;
11     for & byte in buffer . iter () {
12         checksum = checksum . wrapping_add ( byte as u32 );
13     }
14
15     let end = dwt :: get_cycles ();
16
17     // Prevent compiler optimization core :: ptr
18     :: write_volatile (& checksum , checksum );
19
20     end - start
21 }

```

4.4.2 Random Access Pattern Evaluation

Random memory access patterns represent worst-case scenarios for cache performance and MPU overhead:

Cache Miss Impact: Analysis of MPU overhead interaction with cache miss penalties, relevant to automotive applications with irregular data access patterns.

Permission Check Distribution: Evaluation of MPU validation overhead across random memory access patterns that may cross protection boundaries.

4.4.3 Memory Allocation Performance

Dynamic memory allocation performance under MPU protection affects automotive applications requiring runtime resource management:

Table 4.2: Memory Allocation Test Scenarios

Block Size	Iterations	Alignment	Usage Pattern
64 bytes	20	4-byte	Small object allocation
128 bytes	20	8-byte	Medium buffer allocation
256 bytes	20	16-byte	Large buffer allocation
512 bytes	20	32-byte	Packet buffer allocation
1024 bytes	20	64-byte	Frame buffer allocation

4.5 Statistical Analysis Methodology

Rigorous statistical analysis ensures reliable interpretation of performance measurements and identification of significant trends.

4.5.1 Measurement Collection Procedures

Systematic measurement procedures minimize sources of variability:

Warm-up Procedures: Each benchmark includes 10-50 warm-up iterations to stabilize processor state including caches, branch predictors, and memory system state.

Sample Size Determination: Statistical power analysis determines minimum sample sizes (typically 100-1000 samples) required for detecting meaningful performance differences.

Environmental Control: Measurements are conducted under controlled conditions with consistent clock frequencies, power supply, and temperature to minimize external variability sources.

4.5.2 Statistical Measures

Comprehensive statistical analysis includes multiple measures relevant to real-time system analysis:

Central Tendency: Mean and median execution times for typical performance characterization

Variability: Standard deviation and coefficient of variation for timing predictability analysis

Distribution Characteristics: 95th and 99th percentile values for worst-case execution time analysis

Outlier Analysis: Identification and characterization of timing outliers that could impact real-time guarantees

4.5.3 Hypothesis Testing

Statistical significance testing validates performance differences between MPU-enabled and baseline configurations:

Paired t-tests: Compare performance between MPU-enabled and disabled configurations for identical workloads.

Analysis of Variance (ANOVA): Evaluate performance differences across multiple MPU configurations to identify optimal settings.

Confidence Intervals: Establish confidence bounds on performance measurements to support engineering decision-making.

4.6 Validation and Verification Procedures

Comprehensive validation ensures measurement accuracy and relevance to automotive applications.

4.6.1 Measurement Accuracy Validation

Multiple validation techniques verify measurement system accuracy:

Synthetic Validation: Known-duration operations (e.g., controlled loop counts) validate cycle counter accuracy and measurement overhead characterization.

Cross-validation: Independent measurement approaches verify consistency of performance results across different measurement techniques.

Reproducibility Testing: Repeated measurements under identical conditions verify measurement system stability and identify sources of measurement variability.

4.6.2 Automotive Relevance Verification

Validation procedures ensure benchmark relevance to actual automotive ECU operation:

Workload Characterization: Comparison of benchmark characteristics with real automotive application profiles to ensure representative behavior.

Timing Constraint Validation: Verification that measured performance characteristics align with actual automotive timing requirements across different ECU categories.

This comprehensive experimental methodology provides the foundation for reliable MPU performance analysis applicable to real automotive ECU development and deployment decisions.

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